



# MSME Manufacturing India — A Simple, Human Project

Target : A Smart Job-Shop Capacity & Automated Order Dispatch System that helps micro-factories, machine job-shops, and component suppliers turn idle machines into paid work while giving OEMs fast, reliable parts.

## Why Now? The Opportunity

Policies like Make in India plus global supply-chain reshoring have created pockets of intense demand. Hundreds of micro-factories sit near OEMs, but coordination is messy: phone calls, paper logs, and WhatsApp threads. That gap causes delays, lost revenue, and frustrated customers. This project fixes that coordination—fast.



# Problem: The Real Bottleneck (Not the Machines)



## Paper & WhatsApp

Machine availability and orders are tracked informally — prone to mistakes and missed slots.



## Poor Capacity Visibility

Shop owners don't see demand across nearby OEMs; OEMs don't see real machine slots.



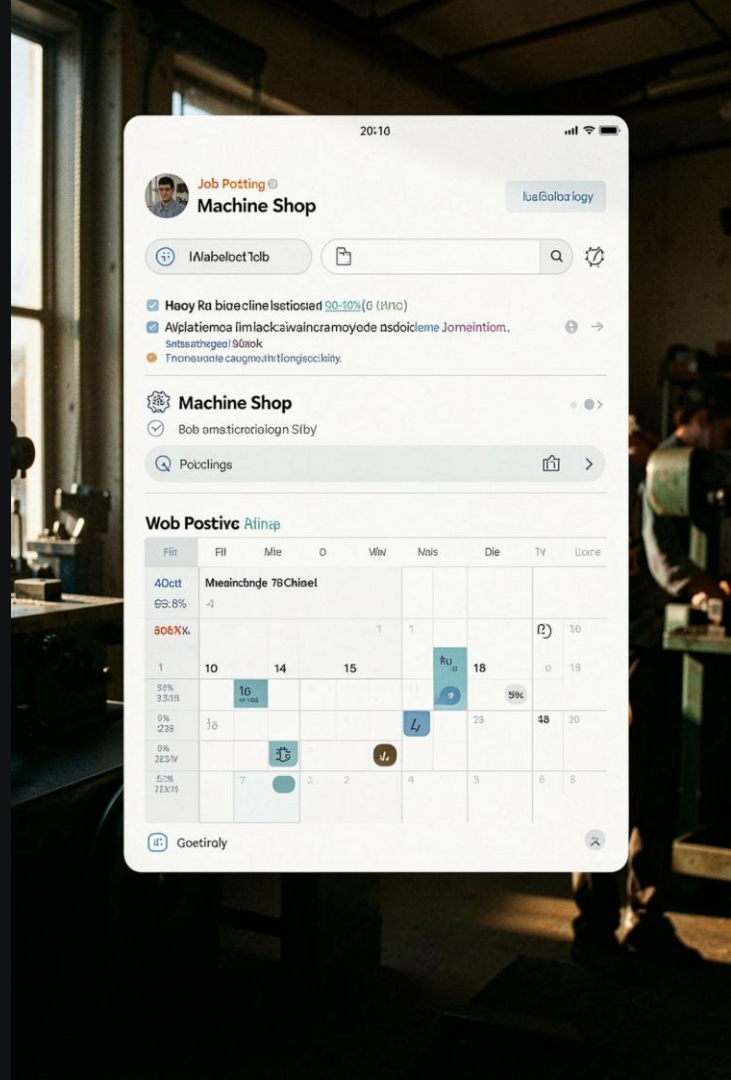
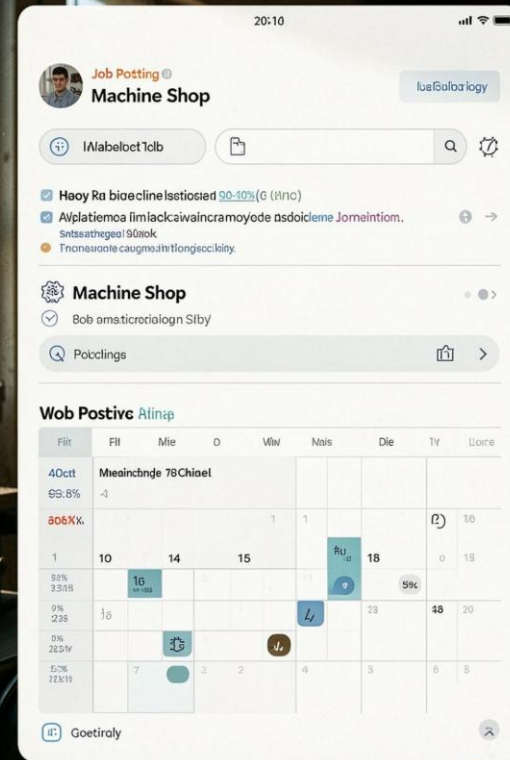
## Order Tracking Gaps

No reliable, auditable proof of quality or stage tracking — disputes and rejections happen after shipping.

Short summary: machines exist, but matching jobs to free machine hours and assuring quality is the hard part.

## Smart Job-Shop Capacity & Automated Order Dispatch System

Local shops post free machine slots on a visual calendar. OEMs post job requirements. A lightweight matching engine pairs them, the order moves through a clear finite-state lifecycle, and visual proof stops rejections before dispatch.



# How It Works — Supply Side (Machine Owners)



## Register Machines

Enter machine type (e.g., Haas CNC Lathe, 3-axis mill), tooling, max part size, and speed. Keep it simple — fields that matter for matching.



## Upload Proof

Before dispatch, upload a part photo and caliper reading. This is a low-code audit step to prevent rejections.

## Mark Idle Slots

Visual calendar: mark free blocks (e.g., Tomorrow 2–8 PM). This becomes supply data the matcher uses in real time.





# How It Works — Demand Side (OEMs / Clients)

## Post Job

Upload part spec: material (e.g., Al 6061), tolerance, quantity, deadline, and a small reference CAD or drawing.

## Auto-Match

System scores nearby shops based on machine fit, available hours, past quality, and distance. Shows top matches with expected delivery time and price estimate.

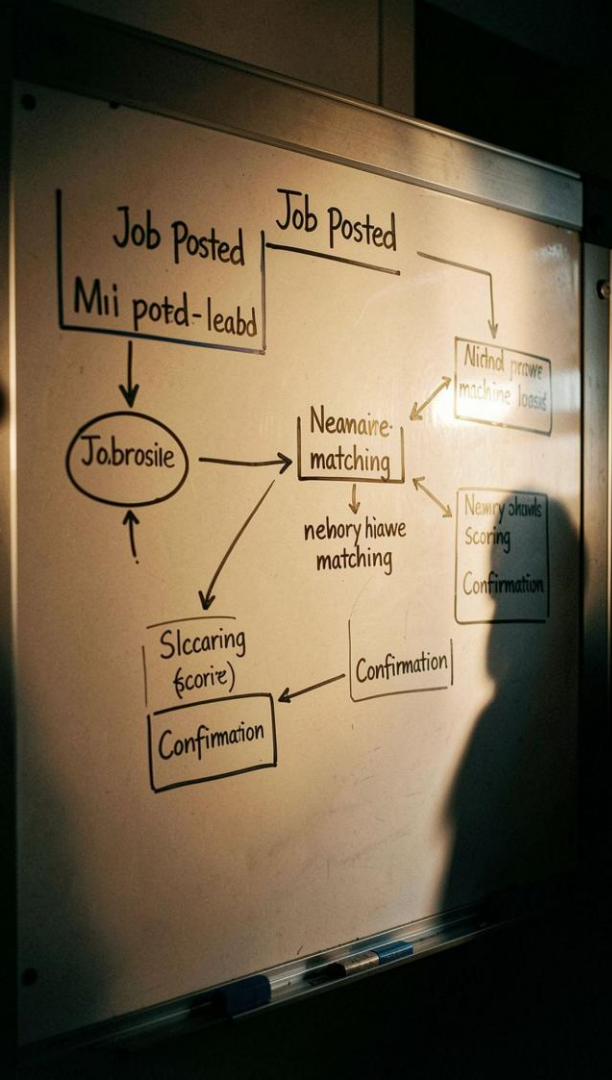
## Approve & Track

Client approves the matched shop. Live updates show order state (Matched → Material Received → Machining → QC → Dispatched).

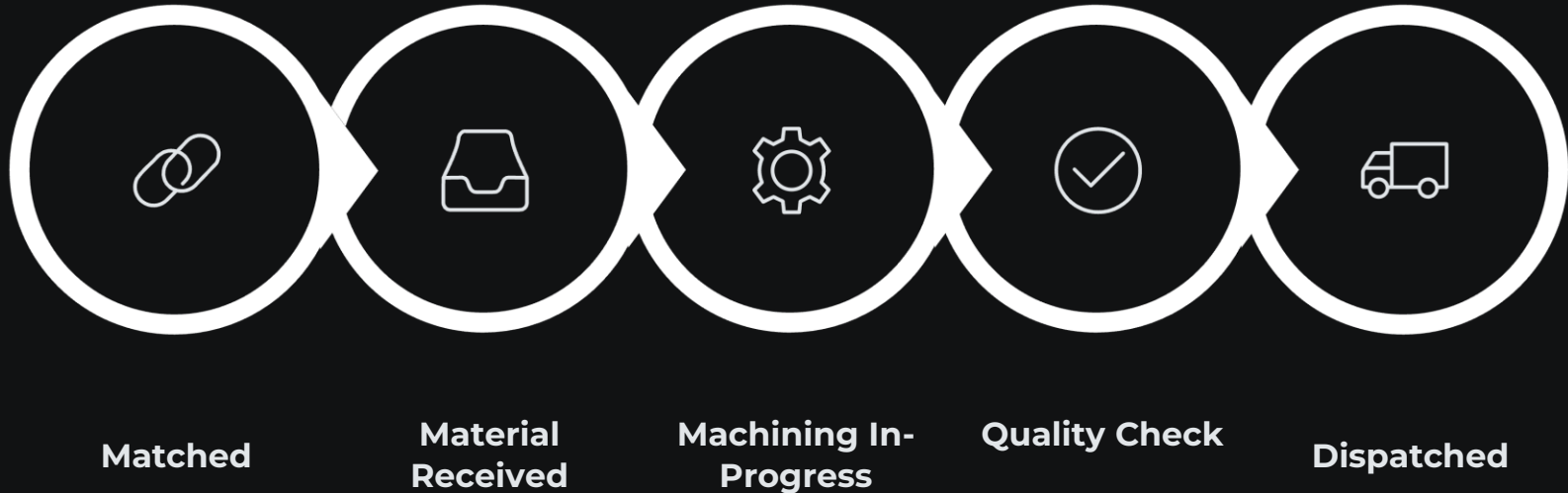


## The Matching Engine — Simple, Effective Logic

Use a weighted scoring function instead of heavy AI. Example factors: machine capability (40%), available hours before deadline (30%), historical quality score (20%), and distance/courier time (10%). The engine ranks options and proposes top 3. No black box — transparent scoring helps trust.



# Finite State Machine (Order Lifecycle)



The FSM enforces steps and prevents skipping. Each transition emits a WebSocket event to the client and the shop owner. If a step stalls (e.g., QC not done in 24 hours), automatic alerts notify both parties and suggest next actions.

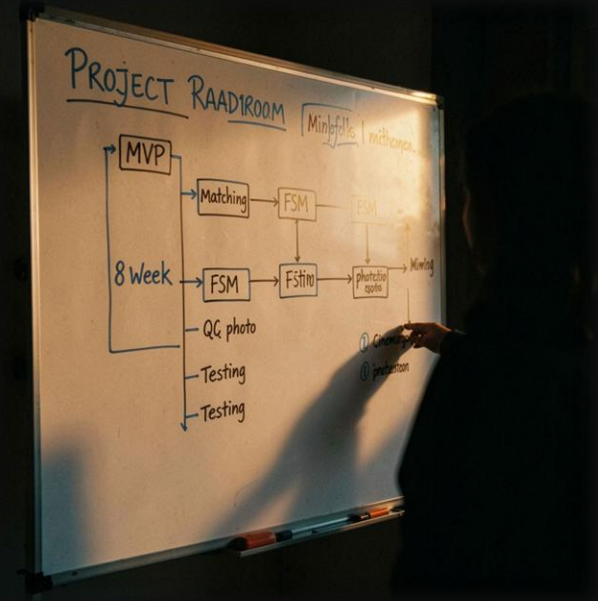


## Simple Tech Stack & 2-3 Month Roadmap



### Core Tech

- Backend: Python (FastAPI) or Node.js (Express)
- DB: PostgreSQL (JSON fields for specs)
- Realtime: WebSockets for live status
- Frontend: React / simple HTML+JS with Tailwind/Bootstrap



### 8-12 Week Plan

1. Week 1-2: Schema, basic UI, machine registration
2. Week 3-4: Slot calendar + job posting
3. Week 5-6: Matching algorithm + FSM
4. Week 7-8: Proof-of-quality upload, WebSockets, testing
5. Week 9-12: Pilot with 5 local shops + iterate



## Impact, Next Steps & Quick Wins



### Increase Shop Revenue

Fill idle hours with matched orders; small shops see immediate utilization gains.



### Faster Turnaround for OEMs

Automated matching reduces sourcing time from days to hours.



### Lower Rejection Risk

Visual proof + caliper readings mean quality approval before shipment.

Next steps: build an MVP, run a week-long pilot with a cluster of 5 shops and 2 OEMs, collect feedback, and iterate. This is a student-scale project with real industrial impact — practical, visible, and resume-worthy.